

University of Wollongong
School of Computing and Information Technology
CSIT121 Object Oriented Design and Programming
Assignment 1

Objectives

- To apply Object Oriented Design (OOD).
- To apply Object Oriented Programming (OOP) using Python.

Note: Please read the submission requirement carefully. Scripts will be used to grade and update your score to Moodle. Failure to name your file name correctly in the required format will result in zero marks updated to Moodle.

Tasks

You are tasked to develop a Personal Training Management System (PTMS) for managing personal training sessions between a Trainer and multiple Trainees. The system should allow the manager to manage trainer and trainee profiles, assign training session, track progress, and calculate overall performance.

Class: Trainee

Represents a person undergoing training.

Attribute	Description
<code>__trainee_id</code>	Unique ID for the trainee
<code>__name</code>	Full name of the trainee
<code>__birthdate</code>	Birthdate of the trainee – date type
<code>__email</code>	Email address

Method	Description
<code>__init__</code>	Initializes the trainee info
<code>get_age()</code>	Returns the age of the Trainee
<code>__str__()</code>	Returns a string showing trainee info

Class: Trainer

Represents a trainer managing multiple trainees.

Attribute	Description
<code>__trainer_id</code>	Unique ID for the trainer
<code>__name</code>	Full name

Method	Description
<code>__init__()</code>	Initialises trainer information
<code>__str__()</code>	Returns a string showing trainer info

Class: ExerciseSession

Represents a single training session.

Attribute	Description
__session_id	Unique ID for the session
__trainer_id	Trainer ID
__trainee_id	Trainee ID
__duration	Duration in minutes
__intensity	Intensity level (1 to 10)
__date	Date of session – date type

Method	Description
__init__()	Initializes the session attributes
__str__()	Returns a formatted string of session info

Class: PersonalTrainingManagementSystem

Attribute	Description
__trainers	List of Trainer objects
__trainees	List of Trainee objects
__sessions	List of Session objects

add_trainer(ID,name)	Add a trainer to the system. Returns True if successful.
add_trainee(ID,name,birthdate,email)	Add a trainee to the system. birthdate -> string (YYYY-MM-DD) Returns True if successful.
create_session(ID,trainer_id,trainee_id,duration,intensity,date)	Create a session and assign a trainer to that trainee. date -> string (YYYY-MM-DD) Returns True if successful.
get_trainer(trainer_id)	Returns the trainer object with that ID.
get_trainee(trainee_id)	Returns the trainee object with that ID.
get_session(session_id)	Returns the session object with that ID.
get_trainee_total_duration(trainee_id)	Returns the total exercise duration of the trainee for all the sessions.
get_trainee_ave_intensity(trainee_id)	Returns the average intensity of the trainee for all the sessions. Round off to two decimal places.
get_trainer_total_duration(trainer_id)	Returns the total duration of the trainer for all the sessions for all trainee.
get_trainer_total_duration_with_trainee(trainer_id,trainee_id)	Returns the total duration of the trainer for all the sessions with that trainee. Returns 0 if that trainee has not been trained under that trainer.
remove_session(session_id)	Remove the session record from the system. Returns True if successful.

Note:

- If an error occur in a method, the method should return a False (unless otherwise stated). Do not throw Exception.
- All ID are passed into the system, do not use an auto-increment variable.
- All instant variables must be private.
- You may implement the variables' getter and setter methods when required.

Test Code and automated grading

The testing code StudentAssertionTest.py has been provided to assist you to test your code.

Ensure that you have passed all the test cases before submission, else zero marks will be awarded if the StudentAssertionTest.py failed to run.

Another script with comprehensive test cases will be used to automate the grading of your assignment.

Submission requirements (very important)

- Submit the python code in a txt file containing the code for Trainee, Trainer, ExerciseSession and PersonalTrainingManagementSystem. Do not include your test code.
- For full time student:
 - File name must be in the form of: **TXXF_NAME_UOWID.txt** where XX is your tutorial group, NAME is your full name (without space or underscore) and UOWID is your 7-digit UOW ID number (**not SIM student ID number**).
 - For example, **T02F_JeffreyTan_8080426.txt**
- For part time student:
 - File name must be in the form of: **TXXP_NAME_UOWID.txt** where XX is your tutorial group, NAME is your full name (without space or underscore) and UOWID is your 7-digit UOW ID number (**not SIM student ID number**).
 - For example, **T02P_JeffreyTan_8080426.txt**
- **Note**
 - Automated script will be used to extract your file for grading. Failure to follow the above instructions will result in failure to extract your submission correctly.
 - Ensure that the UOW ID number is correct as it will be used by the scripts to update your marks in Moodle. Wrong UOW ID number will result in your marks not updated in Moodle.
 - Note as there are many tutorial groups, failure to name your file name correctly in the above format might result in zero marks.
- Late submission will be penalized 5% per day late.
- Late submission more than 4 days late will be awarded a mark of zero